Case Study Rubric

Policy Uncertainty and Housing Market Volatility in Charlottesville

DS 4002 | Spring 2026 | Alison Pike

Overview

Why am I doing this? This case study gives you hands-on experience building a full data science pipeline using real, local data. You will practice scraping text from government documents, transforming that text into a quantitative measure, combining it with numerical market data, and using a time-series regression model to draw conclusions. This mirrors the kind of analysis done in economics, public policy, and urban planning.

What am I going to do? You will collect text from Charlottesville Housing Advisory Committee (HAC) meeting minutes, compute a monthly uncertainty score, merge it with CAAR housing market data, and run an AR(1) time-series regression to test whether uncertainty predicts housing price volatility.

Research Question: Does the level of uncertainty-related language in Charlottesville HAC meeting minutes predict short-term changes in regional housing price volatility in 2025?

Target audience: 2nd year UVA Data Science students with basic Python experience.

Deliverables

Submit a GitHub link or zip file containing all of the following:

1. Written Report (1–2 pages, PDF) — research question, methodology, results, and limitations
2. Python Code — Jupyter notebook or script, documented and reproducible
3. At least two visualizations (e.g., time series plot, scatterplot of uncertainty vs. volatility)

Steps to Follow

1. **Data Collection** — Scrape HAC meeting minutes (.txt or .pdf) from https://www.charlottesville.gov/1077/Agendas-Minutes and monthly CAAR housing reports from https://www.virginiacountryliving.com/caar-market-reports.php. Alternatively, use the pre-cleaned dataset in DATA/.
2. **Preprocessing** — Clean text (remove headers, timestamps, special characters). Exclude months with no meeting or no CAAR report.
3. **Uncertainty Score** — Count sentences containing both housing-related and uncertainty-related synonyms. Compute: Uncertainty Score = (# uncertainty terms) / (# qualifying sentences). Score range: 0–1.
4. **Housing Price Volatility** — Compute month-over-month % change in median sales price: HPV = (Pt - Pt-1) / Pt-1. Score range: -1 to 1.
5. **Merge Datasets** — Join uncertainty scores with CAAR market data by month. Use a one-month lag: uncertainty at time t predicts volatility at time t+1.
6. **Time-Series Regression** — Fit an AR(1) model: Vol_t = α + β1(Uncertainty_t) + β2(Vol_t-1) + u_t. Report coefficients, p-values, R^2, and Adjusted R^2.
7. **Report** — Summarize your process, visualizations, regression results, and interpretation. Discuss limitations honestly — especially the small sample size (n=8).

Assessment Rubric

You will meet expectations on this case study when you satisfy the following criteria:

Category	Criteria
Data Collection	HAC meeting minutes and CAAR reports are collected for all available months of 2025. Months with missing data are excluded and documented. Data sources are cited.
Uncertainty Score	Uncertainty score is computed correctly as (# uncertainty terms) / (# qualifying sentences) per month. Housing-related and uncertainty-related word lists are defined and documented.
Price Volatility	Housing price volatility is computed as month-over-month % change in median sales price. The one-month lag structure is applied correctly.
Regression Model	An AR(1) time-series regression is fitted. Output includes: coefficients for uncertainty and lagged volatility, p-values, R^2, and Adjusted R^2. Results are correctly interpreted.
Visualizations	At least two plots are included and clearly labeled. Suggested: (1) time series of uncertainty score vs. median sales price; (2) scatterplot of uncertainty score vs. price volatility.
Written Report	Report is 1–2 pages. Includes: research question, methodology summary, regression results, interpretation, and at least two limitations (e.g., small sample size, omitted variables).
Code Quality	Code is organized, commented, and reproducible. A second person should be able to run it from start to finish using the provided data and obtain the same results.

Resources Provided

- Pre-cleaned combined dataset (DATA/cleaned_housing_market_data.csv)
- Blog-style explainer on time-series regression (MATERIALS/explainer_timeseries.pdf)
- Reference article on text-based uncertainty measurement (MATERIALS/reference_uncertainty.pdf)
- Original project repo: https://github.com/emilyjmoore/DS_4002_Project_1

A Note on Sample Size

This dataset has only 8 usable observations. This is a real constraint, and part of the learning here is understanding what you can and cannot conclude from a small sample. Your results may not be statistically significant — that is okay. Focus on correctly executing the pipeline, interpreting your coefficients economically, and being transparent about limitations.